Huawei Access Node MAR-ACC-26 Routing Protocol Flap SOP

Node: MAR-ACC-26 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8640

Description

The occurrence of a Routing Protocol Flap on the Huawei Access Node MAR-ACC-26 in the Marathahalli Core site is attributed to the intricate interplay between the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding within the node's Access Role. This complex failure domain is a direct consequence of the underlying protocol mechanics, where the convergence of routing protocols in a high-availability network such as the Marathahalli Core can lead to oscillations and instability. The Huawei Access Node, being a critical component of the network, plays a pivotal role in this scenario, as its ability to maintain stable routing information and adapt to changing network conditions is crucial for preventing such failures. Furthermore, the CLI dialect used on the Huawei equipment, which is optimized for high-speed data transfer and low-latency communication, can sometimes contribute to the complexity of troubleshooting and resolving such issues.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Marathahalli Core site to physically access the node.

- Verify the status of the redundant links connected to the Access Node MAR-ACC-26.

- Check the network topology and ensure that there are no configuration inconsistencies or anomalies that may be contributing to the issue.

Potential Root Causes

Cause	Probability
BGP route churn due to inconsistent routing information	High
OSPF adjacency loss due to network instability	Medium
ISIS LSP flooding due to misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Routing table for process bgp 64.64.64.64, Description: This command displays the BGP routing table, allowing us to verify the routing information and identify any inconsistencies or anomalies.

Command: display ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays the OSPF neighbor list, enabling us to verify the adjacency status and identify any issues that may be contributing to the failure.

Command: display isis lsp, Expected Output: IS-IS LSP information, Description: This command displays the ISIS LSP information, allowing
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us to verify the LSP flooding and identify any misconfigurations that may be contributing to the issue.
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Resolution Steps

Step 1: Step: Verify BGP routing information, Command: display bgp routing-table, Description: Verify the BGP routing table to ensure that the routing information is consistent and accurate.

Step 2: Step: Reset OSPF adjacency, Command: reset ospf neighbor, Description: Reset the OSPF adjacency to ensure that the neighbor list is updated correctly and the adjacency is established properly.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp, Description: Clear the ISIS LSP flooding to prevent further instability and ensure that the LSP information is up-to-date.

Post-Check Verification

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Command: display bgp routing-table, Expected Output: Routing table for process bgp 64.64.64.64, Description: Verify the BGP routing table to ensure that the routing information is consistent and accurate after the resolution steps.

Command: display ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify the OSPF neighbor list to ensure that the adjacency is established properly and the neighbor list is updated correctly.

Command: display isis lsp, Expected Output: IS-IS LSP information, Description: Verify the ISIS LSP information to ensure that the LSP flooding is cleared and the LSP information is up-to-date.
```

Roll-back Procedures

- Step: Restore original BGP configuration, Command: rollback bgp configuration, Description: Restore the original BGP configuration to prevent any further disruptions to the network.

- Step: Reset OSPF adjacency, Command: reset ospf neighbor, Description: Reset the OSPF adjacency to ensure that the neighbor list is updated correctly and the adjacency is established properly.

- Step: Clear ISIS LSP flooding, Command: clear isis lsp, Description: Clear the ISIS LSP flooding to prevent further instability and ensure that the LSP information is up-to-date.